

Case name: NewNMRConfiguration

Description: New Members' Rocket prototype by Anthony Cardoza

Thu Sep 24 21:25:24 2026

Table 1. Propellant Specification

Component	Temperature, F	Pressure, psi	Enthalpy, Btu/lb-mol	Enthalpy, Btu/lbm	Mass fraction
RP-1	77.0	14.6959	-10626.6881	-760.343	0.2915452
O2(L)	-297.4	14.6959	-5579.9487	-174.380	0.7084548
Total			-8028.2104	-345.215	

Propellant exploded formula:

$C_{0.485118} H_{0.945979} O_{1.029765}$ (based on 1 mole)

$C_{19.713319} H_{38.440973} O_{41.845708}$ (by mass %)

α : 0.7135152 (oxidizer excess coefficient)

O/F: 2.4300000

O/F₀: 3.4056740 (stoichiometric)

ρ : 60.82582 lbm/ft³

Table 2. Combustion Properties

Parameter	Injector	Nozzle inlet	Nozzle throat	Nozzle exit Unit
Pressure	360.0000	357.8509	207.3993	13.5809 psi
Temperature	6284.5821	6282.1024	5971.0689	4496.1986 R
Enthalpy	-7792.5966	-7830.3825	-14620.3756	-44404.3224 Btu/lb-mol
	-345.2155	-346.8749	-638.7720	-1844.7997 Btu/lbm
Entropy	0.0636	0.0636	0.0645	0.0678 Btu/(lb-mol·R)
	2.8174	2.8176	2.8176	2.8176 Btu/(lbm·R)
Internal energy	-20273.0069	-20305.8683	-26478.1872	-53333.2221 Btu/lb-mol
	-898.1032	-899.5214	-1156.8461	-2215.7553 Btu/lbm
Specific heat (p=const)	1.5960	1.5964	1.5047	0.7099 Btu/(lbm·R)
Specific heat (V=const)	1.3529	1.3532	1.2860	0.6084 Btu/(lbm·R)
Gamma	1.1798	1.1797	1.1701	1.1669
Isentropic exponent	1.1357	1.1356	1.1333	1.1616
Gas constant	0.0880	0.0880	0.0868	0.0825 Btu/(lbm·R)
Molecular weight	22.5731	22.5741	22.8883	24.0700
Density	0.1205	0.1198	0.0741	0.0068 lbm/ft ³
Sonic velocity	3964.9401	3964.0126	3833.9890	3284.5709 ft/s
Mach number	0.0000	0.0727	1.0000	2.6382
Area ratio	8.2200	8.2200	1.0000	4.8400 A/At
Mass flux	34.5405	34.5405	284.0239	58.7065 lbm/ft ² ·s
Viscosity	0.7225	0.7223	0.6978	0.5740 x10 ⁻⁴ lbm/(ft·s)
Conductivity, frozen	0.3631	0.3629	0.3461	0.2661 Btu/(hr·ft·R)
Specific heat (p=const), frozen	0.4913	0.4913	0.4891	0.4739 Btu/(lbm·R)
Prandtl number, frozen	0.6336	0.6336	0.6390	0.6623
Conductivity, effective	1.6064	1.6064	1.4570	0.5452 Btu/(hr·ft·R)
Specific heat (p=const), effective	1.5960	1.5964	1.5047	0.7099 Btu/(lbm·R)
Prandtl number, effective	0.4652	0.4652	0.4670	0.4843

Table 3. Combustion Products

Product	Injector mass fraction	Injector mole fraction	Nozzle inlet mass fraction	Nozzle inlet mole fraction	Nozzle throat mass fraction	Nozzle throat mole fraction	Nozzle exit mass fraction	Nozzle exit mole fraction
CO	0.4274796	0.3445030	0.4274337	0.3444804	0.4167612	0.3405537	0.3735296	0.3209864
CO2	0.2463378	0.1263505	0.2464102	0.1263929	0.2632003	0.1368840	0.3311533	0.1811169
COOH	0.0000173	0.0000087	0.0000172	0.0000086	0.0000101	0.0000051	0.0000004	0.0000002
H	0.0016631	0.0372466	0.0016628	0.0372410	0.0014162	0.0321581	0.0003537	0.0084465
H2	0.0091419	0.1023681	0.0091414	0.1023669	0.0089170	0.1012436	0.0089863	0.1072986
H2O	0.2476966	0.3103640	0.2477211	0.3104077	0.2559646	0.3252010	0.2804239	0.3746712
H2O2	0.0000087	0.0000058	0.0000087	0.0000057	0.0000049	0.0000033		
HCHO,form aldehy	0.0000005	0.0000004	0.0000005	0.0000004	0.0000003	0.0000002		
HCO	0.0000202	0.0000157	0.0000201	0.0000156	0.0000113	0.0000089	0.0000004	0.0000004
HC00H	0.0000021	0.0000010	0.0000021	0.0000010	0.0000012	0.0000006		
HO2	0.0000657	0.0000449	0.0000655	0.0000448	0.0000376	0.0000261	0.0000004	0.0000003
O	0.0077444	0.0109263	0.0077380	0.0109178	0.0055932	0.0080014	0.0002088	0.0003141
O2	0.0180667	0.0127449	0.0180558	0.0127377	0.0139483	0.0099770	0.0005981	0.0004499
OH	0.0417553	0.0554201	0.0417229	0.0553795	0.0341339	0.0459370	0.0047449	0.0067153
Gaseous fraction:	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000
Condensed fraction:	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000

Table 4. Ideal Performance

Parameter	Sea level	Optimum expansion	Vacuum Units
Characteristic velocity		5854.8964	ft/s
Effective exhaust velocity	8577.4360	8665.4341	9737.2275 ft/s
Specific impulse (by mass)	266.5949	269.3299	302.6423 lbf-s/lbm
Specific impulse (by weight)	266.5949	269.3299	302.6423 s
Thrust coefficient	1.4650	1.4800	1.6631
Thrust	423.8651	428.2137	481.1777 lbf
Altitude	0.0000	0.4104	mi
Ambient pressure	14.6959	13.5809	0.0000 psi

Table 5. Estimated Delivered Performance

Parameter	Sea level	Optimum expansion	Vacuum Units
Characteristic velocity		5704.6061	ft/s
Effective exhaust velocity	8117.8121	8203.5513	9247.8327 ft/s
Specific impulse (by mass)	252.3093	254.9742	287.4314 lbf-s/lbm
Specific impulse (by weight)	252.3093	254.9742	287.4314 s
Thrust coefficient	1.4230	1.4381	1.6211
Thrust	401.1522	405.3891	456.9936 lbf
Altitude	0.0000	0.4104	mi
Ambient pressure	14.6959	13.5809	0.0000 psi

Table 6. Altitude Performance

Altitude, mi	Pressure, psi	Specific impulse, lbf-s/lbm	Specific impulse, s	Thrust coefficient	Thrust, lbf
0.5336	13.2600	8228.2270	255.7411	1.4424	406.6085
0.5415	13.2396	8229.7969	255.7899	1.4427	406.6861
0.5537	13.2081	8232.2159	255.8651	1.4431	406.8056
0.5683	13.1708	8235.0858	255.9543	1.4436	406.9474
0.5844	13.1291	8238.2909	256.0539	1.4441	407.1058
0.6023	13.0839	8241.7677	256.1620	1.4448	407.2776
0.6211	13.0357	8245.4748	256.2772	1.4454	407.4608
0.6412	12.9849	8249.3824	256.3986	1.4461	407.6539
0.6621	12.9317	8253.4679	256.5256	1.4468	407.8558
0.6839	12.8765	8257.7132	256.6576	1.4476	408.0656
0.7068	12.8194	8262.1033	256.7940	1.4483	408.2825
0.7302	12.7606	8266.6258	256.9346	1.4491	408.5060
0.7543	12.7002	8271.2699	257.0789	1.4499	408.7355
0.7792	12.6384	8276.0264	257.2268	1.4508	408.9706
0.8047	12.5752	8280.8869	257.3778	1.4516	409.2108
0.8310	12.5107	8285.8441	257.5319	1.4525	409.4557
0.8578	12.4450	8290.8914	257.6888	1.4534	409.7051
0.8850	12.3783	8296.0228	257.8483	1.4543	409.9587
0.9129	12.3106	8301.2329	258.0102	1.4552	410.2162
0.9413	12.2418	8306.5165	258.1744	1.4561	410.4773
0.9700	12.1722	8311.8690	258.3408	1.4570	410.7418
0.9994	12.1018	8317.2862	258.5092	1.4580	411.0095
1.0291	12.0305	8322.7639	258.6794	1.4590	411.2802
1.0595	11.9586	8328.2985	258.8514	1.4599	411.5537
1.0902	11.8859	8333.8863	259.0251	1.4609	411.8298
1.1213	11.8126	8339.5242	259.2003	1.4619	412.1084
1.1529	11.7386	8345.2089	259.3770	1.4629	412.3893
1.1849	11.6641	8350.9374	259.5551	1.4639	412.6724
1.2173	11.5891	8356.7070	259.7344	1.4649	412.9575
1.2500	11.5136	8362.5150	259.9149	1.4659	413.2445
1.2832	11.4376	8368.3588	260.0965	1.4669	413.5333
1.3167	11.3611	8374.2361	260.2792	1.4680	413.8237
1.3505	11.2843	8380.1444	260.4628	1.4690	414.1157
1.3847	11.2071	8386.0815	260.6474	1.4701	414.4091
1.4194	11.1295	8392.0455	260.8327	1.4711	414.7038
1.4542	11.0517	8398.0341	261.0189	1.4721	414.9997
1.4895	10.9735	8404.0454	261.2057	1.4732	415.2968
1.5250	10.8950	8410.0776	261.3932	1.4743	415.5949
1.5609	10.8163	8416.1289	261.5813	1.4753	415.8939
1.5972	10.7374	8422.1974	261.7699	1.4764	416.1938
1.6337	10.6583	8428.2816	261.9590	1.4775	416.4945
1.6705	10.5790	8434.3797	262.1485	1.4785	416.7958
1.7077	10.4995	8440.4903	262.3385	1.4796	417.0978
1.7450	10.4199	8446.6118	262.5287	1.4807	417.4003
1.7828	10.3402	8452.7428	262.7193	1.4817	417.7032
1.8208	10.2603	8458.8817	262.9101	1.4828	418.0066
1.8591	10.1804	8465.0274	263.1011	1.4839	418.3103
1.8978	10.1000	8471.2099	263.2933	1.4850	418.6158

Table 7. Throttled Performance (delivered)

Throttle value	Mass flow rate, lbm/s	Pressure, psi	Specific impulse, lbf-s/lbm Sea level	Specific impulse, s Sea level	Thrust, lbf Sea level	Specific impulse, lbf-s/lbm Optimum expansion	Specific impulse, s Optimum expansion	Thrust, lbf Optimum expansion	Specific impulse, lbf-s/lbm Vacuum	Specific impulse, s Vacuum	Thrust, lbf Vacuum
1.0000	1.5899	360.000	252.3093	252.3093	401.152	254.9742	254.9742	405.3891	287.4314	287.4314	456.9936
		0			2						

Table 8. Chamber Size

Combustion chamber size				Nozzle size	
Dc	2.87	in	Type	Conical nozzle	
Dt	1.00	in	Rn	0.50	in
Lcyl	3.52	in	Tn	15.00	deg
Lc	4.35	in	Te	15.00	deg
L*	31.41	in	De	2.20	in
R1	0.50	in	Le	2.31	in
R2	0.00	in	Le/Dt	2.31	
b	60.00	deg	Le/Lc15		%
Ac/At	8.22		Ae/At	4.84	

Parameter		Engine	Chamber
Thrust	sea level	401.1522	401.1522 lbf
	opt exp	405.3891	405.3891 lbf
	vacuum	456.9936	456.9936 lbf
Specific Impulse	sea level	252.3093	252.3093 lbf-s/lbm
	opt exp	254.9742	254.9742 lbf-s/lbm
	vacuum	287.4314	287.4314 lbf-s/lbm
Mass flow rate	total	0.7212	0.7212 lbm/s
	oxidizer	0.5109	0.5109 lbm/s
	fuel	0.2103	0.2103 lbm/s
	film coolant	n.a.	n.a. lbm/s
Number of chambers		1	

Table 10. Thermal Analysis